Airflow ML Pipeline: Churn Feature Selection with Cuckoo Search

An end-to-end Airflow-based machine learning pipeline project for customer churn prediction

Project Summary: An end-to-end Airflow-based machine learning pipeline was developed for customer churn prediction. Churn prediction was performed using a Random Forest model, and feature selection was applied using the Cuckoo Search algorithm. In the optimized model, the number of features was reduced from 30 to 12, while the F1-score improved from 0.553 to 0.589 and recall improved from 0.492 to 0.684.

1. Project Objective

The objective of this project is to develop an end-to-end machine learning pipeline that predicts the risk of customer churn. The project also combines optimization techniques studied in a master's program with a real-world data science problem.

In this project, the Cuckoo Search algorithm was not used directly as a prediction model. Instead, it was used for **feature selection**. This allowed the project to test whether the model could achieve better performance with fewer variables.

2. Technologies Used

Area	Technology Used
Workflow orchestration	Apache Airflow
Programming language	Python
Data processing	Pandas, NumPy
Machine learning	Scikit-learn
Model	Random Forest Classifier
Optimization	Cuckoo Search Algorithm
Output formats	CSV, PKL model file
Execution environment	Airflow on Docker

3. Dataset

The project uses the **Telco Customer Churn** dataset. The dataset contains telecom customers' subscription information, service usage details, payment information, and churn status.

Example Columns

- `gender`
- `SeniorCitizen`
- `Partner`
- `Dependents`
- `tenure`
- `InternetService`
- `Contract`
- `PaymentMethod`
- `MonthlyCharges`
- `TotalCharges`
- `Churn`

4. Airflow DAG Structure

The Airflow DAG was designed to be triggered manually. Since the dataset used in this project is static, a daily schedule is not required. However, the structure can easily be extended to run daily, weekly, or monthly in a real production environment when new data becomes available.

```
check_dataset_exists
        ↓
validate_columns
        ↓
clean_churn_data
        ↓
train_baseline_model
        ↓
cuckoo_search_feature_selection
        ↓
train_optimized_model
        ↓
generate_churn_predictions
```

5. Pipeline Steps

5.1 Dataset Check

The `check_dataset_exists` task checks whether the raw data file exists in the correct folder. If the file does not exist, the pipeline fails.

5.2 Column Validation

The `validate_columns` task checks whether the expected columns are available in the dataset. If any required column is missing, the process is stopped.

5.3 Data Cleaning and Encoding

The following operations are performed in the `clean_churn_data` task:

- The `customerID` column is removed because it is not meaningful for the model.
- The `TotalCharges` column is converted to a numeric format.
- Records with missing values are removed.
- The target variable `Churn` is converted as `Yes=1` and `No=0`.
- Categorical variables are converted into numeric format using one-hot encoding.

5.4 Baseline Model Training

In the `train_baseline_model` task, a Random Forest model is trained using all available features. This model serves as the starting point for comparison with the optimized model after Cuckoo Search.

5.5 Feature Selection with Cuckoo Search

In the `cuckoo_search_feature_selection` task, the Cuckoo Search algorithm is used to search for the best feature subset. Each solution is represented as a feature selection combination:

```
[1, 0, 1, 1, 0, 1, ...]
```

Where:

- `1`: The feature is included in the model.
- `0`: The feature is excluded from the model.

The objective of the optimization is to find the feature combination that improves the F1-score. A small feature penalty is applied to prevent the selection of too many features.

5.6 Optimized Model Training

The `train_optimized_model` task trains a new Random Forest model using the features selected by Cuckoo Search.

5.7 Generating Churn Predictions

The `generate_churn_predictions` task uses the optimized model to generate churn predictions and churn probabilities for all customers. Customers are also segmented into `low`, `medium`, and `high` risk groups.

6. Model Performance Comparison

Model	Accuracy	Precision	Recall	F1-score	ROC-AUC	Feature Count
Baseline Random Forest	0.789	0.632	0.492	0.553	0.818	30
Cuckoo Optimized Random Forest	0.746	0.516	**0.684**	**0.589**	0.792	**12**

7. Interpretation of Results

The baseline model produced an F1-score of 0.553 and a recall of 0.492 using 30 features. The model optimized with Cuckoo Search produced an F1-score of 0.589 and a recall of 0.684 using only 12 features.

This result shows that feature selection with Cuckoo Search provided a significant improvement, especially in terms of **recall**.

> **Business Interpretation:** In churn problems, recall is a critical metric because the goal is to identify as many customers at risk of leaving as possible. Since the optimized model can capture more churn customers, it may be more useful for customer retention campaigns.

8. Generated Files

File	Description
`telco_churn_processed.csv`	Cleaned and model-ready dataset
`baseline_churn_model.pkl`	Baseline model trained with all features
`baseline_metrics.csv`	Baseline model performance metrics
`selected_features.csv`	Feature list selected by Cuckoo Search
`cuckoo_feature_selection_metrics.csv`	Cuckoo Search optimization results
`optimized_churn_model.pkl`	Optimized model trained with selected features
	Optimized model performance metrics

`optimized_metrics.csv`	
`model_comparison.csv`	Comparison between baseline and optimized models
`churn_predictions.csv`	Customer-level churn predictions and risk segments

9. Prediction Output

The prediction file generated by the final model contains the following columns:

Column	Description
`customer_row_id`	Customer row ID
`churn_prediction`	1 = churn risk exists, 0 = no churn risk
`churn_probability`	Churn probability calculated by the model
`risk_segment`	Low, medium, or high risk segment

Example Prediction Output

```
customer_row_id,churn_prediction,churn_probability,risk_segment
1,1,0.753,high
2,0,0.013,low
3,1,0.887,high
4,0,0.000,low
5,1,0.900,high
```

10. Project Strengths

- An end-to-end ML pipeline was created.
- A task-based, traceable, and rerunnable structure was built with Airflow.
- The Cuckoo Search algorithm was applied to a real ML optimization problem.
- Baseline and optimized model performances were compared.
- The number of features was reduced from 30 to 12 while improving F1-score and recall.
- The final prediction output was converted into risk segments that can be understood by business teams.

11. Future Improvements

- The Cuckoo Search algorithm can also be used for hyperparameter optimization.
- MLflow can be added for model versioning.
- A Power BI dashboard can be created to visualize churn risk segments.

- The DAG can be scheduled when real-time or periodic new customer data becomes available.
- Model explainability can be improved with SHAP analysis.
- A campaign recommendation system can be added for high-risk customers.